

Problem Set-5

 **Problem 1.** A large population of ALOHA users manages to generate 50 requests/sec, including both originals and retransmissions. Time is slotted in units of 40 msec. $G = \lambda * t$

- What is the chance of success on the first attempt?
- What is the probability of exactly k collisions and then a success?
- What is the expected number of transmission attempts needed?

Problem 2. Recall that with the CSMA/CD protocol, the adapter waits $K * 512$ bit times after a collision, where K is drawn randomly. For $K=100$, how long does the adapter wait until returning to Step 2 for a 10 Mbps Ethernet? For a 100 Mbps Ethernet?

Problem 3. Consider building a CSMA/CD network running at 1 Gbps over a 1-km cable with no repeaters. The signal speed in the cable is 200,000 km/sec. What is the minimum frame size?
use twice for csma cd

Problem 4. Suppose nodes A and B are on the same 10 Mbps Ethernet segment, and the propagation delay between the two nodes is 225 bit times. Suppose A and B send frames at the same time, the frames collide, and then A and B choose different values of K in the CSMA/CD algorithm. Assuming no other nodes are active, can the retransmissions from A and B collide? For our purposes, it suffices to work out the following example. Suppose A and B begin transmission at $t=0$ bit times. They both detect collisions at $t=225$ bit times. They finish transmitting jam signal at $t=225+48=273$ bit times. Suppose $K_A=0$ and $K_B=1$. At what time does B schedule its retransmission? At what time does A begin transmission? (Note, the nodes must wait for an idle channel after returning to Step 2-- see protocol.) At what time does A's signal reach B? Does B refrain from transmitting at its scheduled time?

Problem 5. Consider a 100Mbps 100BT Ethernet. In order to have an efficiency of .50, what should be the maximum distance between a node and the hub? Assume a frame length of 64 bytes and that there are no repeaters. Does this maximum distance also ensure that a transmitting node A will be able to detect whether any other node transmitted while A was transmitting? Why or why not? How does your maximum distance compare to the actual 100 Mbps standard?